



WORKFLOW

REPLICATION OF TAHOE-100M DATA ANALYSIS FOR LNCRNA

1) CREATION OF THE MERGED GENESETS

1. gene_vocabulary.jsonl
2. GO_Biological_Process_2021_lncRNA.gmt
3. Reactome_2022_lncRNA.gmt
4. c5.go.v2023.1.Hs.symbols.gmt
c2.cp.reactome.v2023.1.Hs.symbols.gmt

merging_genesets.ipynb

1. Keeping only the genes with gene symbols
2. Filtering of genes to the intersection of the reference genome and the gene set collections
3. Merging the lnc genesets and the msigdb genesets

1. lncrna_sets.txt
2. msigdb_sets.txt

Summary:
lncRNA genes: 3290
lncRNA sets: 7743
msigdb genes: 19396
all unique genes 22607

1) CREATION OF THE MERGED GENESETS

Merging_genesets02.ipynb

1. Merging lnc genesets and msigdb genesets
2. Extracting all unique genes
3. Filtering genesets so that they have 10 – 500 genes. (I didn't use the result of this (output 3.) because I have a similar step later.

1. enriched_lncrna_sets.txt
2. all_unique_genes.txt
3. filtered_enriched_sets.txt

Successfully Enriched: 5901
No Match (Unchanged):
1784 Failed/Mismatched: 0

Found 21115 unique
genes.

Sets kept (10-500 genes): 6182
Sets removed: 1561

2) HIGH VARIABLE GENES

1. adata.h5ad (Tahoe model)
2. all_unique_genes.txt

HVG_scvi02.py

1. Generating model-denoised expression of 10.000 random cells
2. Calculating standard deviations for each gene and sorting
3. The highest 50% sorted genes were saved.

scvi_HVGs.txt

Found 21046 valid genes
out of 21115 in the list.

Identified 10523 Highly Variable
Genes (Top 50%).

Top 5 most variable:

	gene	std_dev
19093	TPRG1L	1846.031250
16206	SERPINB9	752.256958
16194	SERPINB1	601.587830
18672	TIPARP	336.144073
4074	DNAH8-DT	202.578476

3) FILTERING WITH HIGH VARIABLE GENES

1. enriched_lncrna_sets.txt
2. scvi_HVGs.txt

Filtering_genesets_HVGs.ipynb

FILTER 1: Total gene count (15 to 500)
FILTER 2: HVG intersection (at least 10)
Sorting by number of HVGs

1. scvi_hvgs_genesets_filtered_15-500-10
2. scvi_hvgs_genesets_filtered_15-500-10_sorted
(the names are different in the script because I changed them later)

Total sets processed: 7743
Sets retained: 4527
Sets discarded: 3216

Max HVGs found: 270
Min HVGs found: 10

4) DIFFERENTIAL VISION SCORES

1. adata.h5ad
2. scvi_hvgs_genesets_filtered_15-500-10_sorted.txt

diff_vision_scores_all40.py

Separation by groups: cell line per plate (674 groups)

FOR each of 100 MC samples:

1. For every group, subsample 100 cells per drug + 100 DMSO cells
2. Concatenate ALL sampled cells across ALL groups into one batch
3. ONE model call for the entire batch
4. Compute Vision scores for the entire batch at once
5. Slice back to each group, compute drug mean - DMSO mean
6. Store differential scores for this sample

AFTER 100 samples:

Median across samples → final scores per group

674

differential_vision_scores.parquet

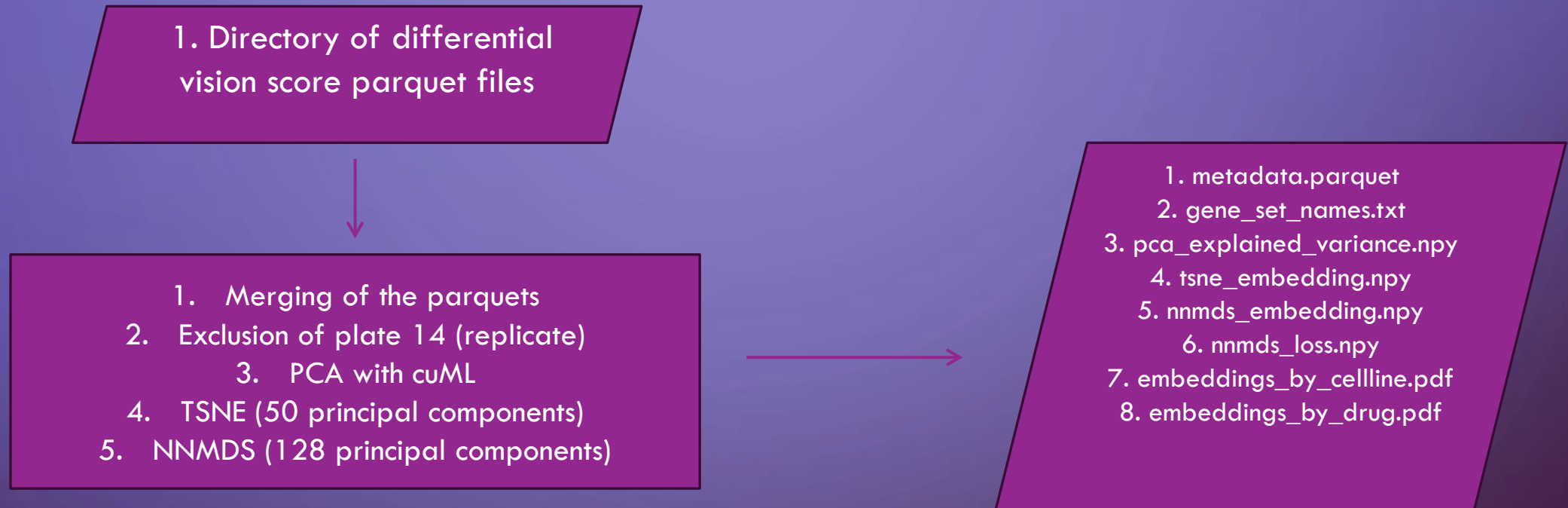
X 17 JOBS

□ Vision score = mean(
 $\log(\text{expr}(\text{gene}) + \epsilon)$ for all genes
in the set)

□ Differential Score =
mean(scores of drug cells) -
mean(scores of DMSO cells)

I don't have them uploaded on git. Around 6MB each.

5) DIMENSIONAL REDUCTION



6) E-DISTANCE CALCULATION

data.h5ad

E_distance.py

Computes E-distance for each (drug, cell_line) comparison within a plate, using the scVI 10-dimensional latent representation.

E-distance formula (Peidli et al. 2024):

$$E = (2 * \text{mean_cross} - \text{mean_treat} - \text{mean_ctrl}) * \text{adjustment}$$

where:

mean_cross = mean pairwise distance between treated and control cells

mean_treat = mean pairwise distance within treated cells

mean_ctrl = mean pairwise distance within control cells

$$\text{adjustment} = (n_{\text{treat}} * n_{\text{ctrl}}) / (n_{\text{treat}} + n_{\text{ctrl}})$$

7) FIGURE 5

1. E-distance parquets (per plate)
2. tsne_embedding.npy
3. nnmds_embedding.npy
4. metadata.parquet



drug_metadata.parquet

Figure5.py

1. Merge E-distance files
2. Match E-distances to embeddings
3. Plot points with E-distance.

Figure5moa.py

1. Merge MOA into metadata
2. Plot figure 5 C

1. e_distance_all.parquet (merged E-distances)
2. figure5_bc.pdf (t-SNE and NNMDS colored by E-distance)

figure5_de_moa.pdf

7) FIGURE 3 FOR LNC GENES

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- ```
graph TD; A[1. adata.h5ad
2. lncrna_sets.txt] --> B[1. Load lncRNA gene sets → extract all unique genes
2. Subsample 200,000 cells (equal from top 47 cell lines)
3. get_normalized_expression from SCVI model
4. PCA on the normalized expression
5. cuML t-SNE on the PCs]; B --> C[1. figure3_lncrna_abc
2. figure3_lncrna_cellline_legend];
```
1. adata.h5ad
  2. lncrna\_sets.txt

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1. figure3\_lncrna\_abc
2. figure3\_lncrna\_cellline\_legend